De Rham-Hodge Theory

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Introduction

In this course we have dealt with the following several important results:
1. The Existence of Mayer-Vietoris Sequence
2. Poincaré Lemmas and the Homotopy Equivalence between Cohomology Classes
3. Poincaré Duality and Poincaré Dual
4. Kunneth Formula
5. The Homotopy Equivalence between Fiber Bundles
6. Thom Isomorphism and Thom Class
7. Hodge Decomposition

During the process we have the following virtuosic techinques:
I. Snake Lemma/Zigzag Lemma. This special technique linked cohomology classes of different orders by constructing a connecting homomorphism.
II. A long exact Mayer-Vietoris Sequence. This is just the zigzag lemma applied to the M-V sequence.
III. Integration Along the Fibers. This notion arises from the fact that integration and exterior derivative are "inverses" of each other along the fiber. In this way we show that trivial fibers do not add to extra cohomology classes.
IV. Five Lemma. This is of utter importance. This is used to prove the isomorphic properties of a map in commutative diagrams. We always use induction on the cardinality of good covers. The long M-V sequence takes care of the inductive process. So the key to use this is to construct a commutative diagram.

The whole process is listed as follows:

$$\begin{tikzcd}
 & & 5 \arrow[dr] & \\
\xrightarrow{\text{III}} & 2 \arrow[r, "\text{II,IV}"] & 3,4 \arrow[r, "\text{II,III,IV}"] & 6
\end{tikzcd}$$

Definition 1. *Given topological space X, Y, their cohomologies $H^*(X), H^*(Y)$ are an abelian groups satisfying functionality: For all $f : X \to Y$, there exists a group homomorphism*

$$f^* : H^*(Y) \to H^*(X). \tag{1}$$

Here $H^*(X)$ is a contravariant functor. In particular, $(f^{-1})^*$(if exists) $= (f^*)^{-1}$.

Definition 2. *A differential complex is a set of indexed vector spaces with linear transformations between each two:*

$$\cdots \xrightarrow{d_{-2}} V_{-1} \xrightarrow{d_{-1}} V_0 \xrightarrow{d_0} V_1 \xrightarrow{d_1} V_2 \xrightarrow{d_2} \cdots \tag{2}$$

where

$$d_i d_{i+1} = 0. \tag{3}$$

Definition 3. *The $k-$the cohomology of a differential complex (V_*, d_*), denoted by $H^k(V_*, d_*)$ is defined as*

$$H^k(V_*, d_*) := \mathrm{ker} d_k / \mathrm{im} d_{k-1}. \tag{4}$$

Comment 1. *It is straightforward to verify that it is indeed a cohomology. f^* are simply pullbacks. Vectors spaces are abelian groups(+), so the quotient group does exist and is abelian.*

Definition 4. *A sequence of vector spaces in a chain is exact if all $H^k(V_*, d_*) = 0$. A short exact sequence is defined as*

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0. \tag{5}$$

So it necessarily means that $\mathrm{im} f = \mathrm{ker} g$. The image from 0 to A is $\{0\}$ so $\mathrm{ker} f = \{0\}$—f is injective. The image of g equals the kernal of $C \to \{0\}$, which is all C so g is surjective.

Definition 5. *The chain maps between two chains (A_*, d_A) and (B_*, d_B): $f_i : A_i \to B_i$ are linear maps with $d_A \circ f_i = f_i \circ d_B$.*

Comment 2. *Chain maps f induces a well defined map $f^* : H^k(A_*, d_A) \to H^k(B_*, d_B)$ since*

$$f^*([\omega]) = f(\omega + d_A \eta) = f\omega + f d_A \eta = f\omega + d_B f\eta = [f\omega]. \tag{6}$$

Theorem 1 (The snake lemma)**.** *Given $0 \to A_i \xrightarrow{f} B_i \xrightarrow{g} C_i \to 0$ consisting of three chains where each three spaces in the chains form a short exact sequence. Then there exists a long exact sequence given by*

$$\cdots H^{i-1}(C) \xrightarrow{d^*} H^i(A) \xrightarrow{f^*} H^i(B) \xrightarrow{g^*} H^i(C) \xrightarrow{d^*} H^{i+1}(A) \cdots \tag{7}$$

Proof. Very obviously, since $\mathrm{im} f = \mathrm{ker} g$,

$$\mathrm{im} f^* = \mathrm{ker} g^*, \tag{8}$$

for we only have to add [...] to each element in the spaces. Now, since g is surjective we have for any closed $c \in C_i$, $c = g(b)$ for some $b \in B_i$ while b might not be closed. And since $0 = dc = d(g(b)) = g(db)$, $db \in \mathrm{ker} g_{i+1} = \mathrm{im} f_{i+1} \Rightarrow db = f_{i+1}(a), a \in A_{i+1}$. Since $0 = ddb = df_{i+1}(a) = f_{i+1}(da)$, $da = 0$ and a is closed. We define $d^*[c] = [a]$. Since $f^*_{i+1}[a] = 0$, d^*

First, we verify that d^* is well-defined. Let $g(b) = g(b') = c$ and from the injectivity of f, $f(a) = b$ and $f(a') = b'$.

$$g(b) - g(b') = g(b - b') = g(\omega) = 0; \quad \omega := b - b', \tag{9}$$

we have $\omega = f(\eta)$ due to exactness. Due to commuativity $f(d\eta) = db - db' = f(a - a')$ and due to injectivity $a - a' = d\eta$. So $[a] = [a']$. Now let $c = d\omega$ then $g(b_{i-1}) = \omega$ and take $b = db_{i-1}$ and then $db = 0$ and $f(a) = 0$ implies $a = 0$.

Now, we check the exactness of the cohomology sequence. First we show that $\mathrm{im} d^* = \mathrm{ker} f^*$:

$$\mathrm{im} d^* = \{[a] : f(a) = db \text{ under the condition that } c = g(b) \text{ is closed.}\} \tag{10}$$

$$\mathrm{ker} f^* = \{[a] : f(a) = db \text{ for some } b \in B_{i+1}.\} \tag{11}$$

Thus we only need to check if $g(b)$ is always closed: $dg(b) = g(db) = g(f(a)) = 0$ since $\mathrm{im} f = \ker g$. Next we show that $\mathrm{im} g^* = \ker d^*$: Obviously $d^*[g^*[b]] = 0$ since $f(a) = db = 0$ and f is injective. Take $c \in \ker d^*$, we have $g(b) = c$ and $f(a) = db$ with a exact: $a = d\omega$. Then let $b' = f(\omega)$. Since $db = f(a) = f(d\omega) = df(\omega) = db'$, $db - db' = 0$ implies b and b' differ by a closed form η. Now

$$g(\eta) = g(b) - g(b') = c - g(f(\omega)) = c. \tag{12}$$

So $c = g(\eta)$, where η is closed. □

Definition 6. *A functor $F : C_1 \to C_2$ maps a category to another. This implies*

$$F : \mathrm{Ob}(C_1) \to \mathrm{Ob}(C_2); \quad \mathrm{Hom}(C_1) \to \mathrm{Hom}(C_2). \tag{13}$$

It must satisfy:

$$F(1_A) = 1_{F(A)}. \tag{14}$$

F is covariant if:

$$F(f) \in \mathrm{Mor}(F(A), F(B)); \quad f \in \mathrm{Mor}(A, B), \tag{15}$$

and that

$$F(g \circ f) = F(g) \circ F(f). \tag{16}$$

F is contravariant if

$$F(f) \in \mathrm{Mor}(F(B), F(A)); \quad f \in \mathrm{Mor}(A, B), \tag{17}$$

and that

$$F(g \circ f) = F(f) \circ F(g). \tag{18}$$

Example 1. *Ω^* is a functor mapping smooth manifolds to vector spaces, and it maps smooth functions to their corresponding pullbacks. H^* is almost the same.*

Proposition 1. *If M_1 is diffeomorphic to M_2, $H^*(M_1) \simeq H^*(M_2)$.*

Proof. Suppose the diffeomorphism is $\phi : M_1 \to M_2$ and its inverse ϕ^{-1}. We have

$$H^*(\phi) : H^*(M_2) \to H^*(M_1), \tag{19}$$

and since

$$H^*(\phi^{-1}) \circ H^*(\phi) = H(\phi \circ \phi^{-1}) = H(1_{M_2}) = 1_{H^*(M_2)}; \tag{20}$$
$$H^*(\phi) \circ H^*(\phi^{-1}) = H(\phi^{-1} \circ \phi) = H(1_{M_1}) = 1_{H^*(M_1)}; \tag{21}$$
$$\tag{22}$$

So $H^*(\phi)$is an inverible linear map(hence an isomorphism) between the two vector spaces. □

Definition 7. *Let $M = U \cup V$, U, V open. Then we have*

$$U \cap V \xrightarrow[\partial_V]{\partial_U} U \coprod V \xrightarrow{(i_U, i_V)} U \cup V = M \tag{23}$$

where ∂_j, i_j are inclusion maps. Applying the contravariant functor Ω^* we have

$$\Omega^*(M) \xrightarrow{(i_U^*, i_V^*)} \Omega^*(U) \oplus \Omega^*(V) \xrightarrow{\partial_V - \partial_U} \Omega^*(U \cap V). \tag{24}$$

This is defined as the Mayer-Vietoris sequence. We should see why $\Omega^*(U \coprod V) = \Omega^*(U) \oplus \Omega^*(V)$: The former one the space of all forms on $U \coprod V$. They include forms on U and forms on V while they have no overlapping. So they are L.I. and the vector space is simpy the direct product.

Proposition 2. *The M-V sequence is exact.*

Proof. To see injectiveness of (i_U^*, i_V^*): For a form $g_I dx^I$ in $\Omega^*(M)$ (or we may say defined on M),

$$(i_U^*, i_V^*) g_I dx^I = \left((g_I \circ i_U) d(x \circ i_U)^I, (g_I \circ i_V) d(x \circ i_V)^I \right) = \left(g_I dx^I \Big|_U, g_I dx^I \Big|_V \right) = (0, 0). \tag{25}$$

We have $g_I(i_U) = g_I(i_V) = 0$ for all I. Since (i_U, i_V) is onto, $g_I(p)$ at any $p \in M$ equals $g_I \circ (i_U)(u)$ or $g_I(i_V)(v)$ for some (u, v). And they are 0. (It seems that the pullback of a surjective map is injective.)

To see $\ker(\partial_V^* - \partial_U^*) = \mathrm{im}((i_U^*, i_V^*))$:

$$(\partial_V^* - \partial_U^*)(i_U^*, i_V^*) f = f \circ (i_U, i_V) \circ (\partial_V - \partial_U) = f \circ (i_U, i_V) \circ \partial_V - f \circ (i_U, i_V) \circ \partial_U = 0 \tag{26}$$

since $(i_U, i_V) \circ \partial_V = (i_U, i_V) \circ \partial_U$. Now suppose $(\partial_V^* - \partial_U^*)(g_I du^I, h_J dv^J) = 0$,

$$\partial_V^*(g_I du^I) = \partial_U^*(h_J dv^J) \Rightarrow (g_I dx^I)\big|_{U \cap V} = (h_J dx^J)\big|_{U \cap V}. \tag{27}$$

This indicates that the two forms agree on $U \cap V$ and by pasting lemma, there exists a set of smooth functions f_I(which are obtained just by pasting g and h together) on M such that

$$f_I dx^I \Big|_U = g_I du^I; \quad f_I dx^I \Big|_V = h_I dv^I \Rightarrow (g_I du^I, h_I dv^I) = (i_U^*, i_V^*) g_I dx^I. \tag{28}$$

To see surjectiveness, let ρ_U and ρ_V be the partition of unity subordinate to U and V. For any form ω on $\Omega^*(U \cap V)$, we have

$$\omega = \rho_V \omega - (-\rho_U \omega) = (\partial_1^* - \partial_0^*)(-\rho_U \omega, \rho_V \omega). \tag{29}$$

□

Corollary 1. *We have a long exact sequence*

$$\cdots H^i(M) \xrightarrow{(i_U^*, i_V^*)} H^i(U) \oplus H^i(V) \xrightarrow{\partial_V^* - \partial_U^*} H^i(U \cap V) \xrightarrow{d^*} H^{i+1}(M) \cdots \tag{30}$$

Additionally,

$$d^*[\omega] \tag{31}$$

Proof. The sequence is simply obtained by Snake lemma. Now take ω closed in $U \cap V$. Then some form in the preimage is $(-\rho_V \omega, \rho_U \omega)$. And then consider the solution to

$$(i_U^*, i_V^*)\eta = (-d\rho_V \wedge \omega - \rho_V d\omega, d\rho_U \wedge \omega + \rho_U d\omega) = (-d\rho_V \wedge \omega, d\rho_U \wedge \omega). \tag{32}$$

Again, since on $U \cap V$ we have

$$d\rho_V \wedge \omega + d\rho_U \wedge \omega = d(1) \wedge \omega = 0, \tag{33}$$

$-d\rho_V \wedge \omega$ and $d\rho_U \wedge \omega$ agree on $U \cap V$ and by pasting lemma we have

$$d^*[\omega] = [\eta], \quad \eta = \begin{cases} -d\rho_V \wedge \omega \text{ on } U; \\ d\rho_U \wedge \omega \text{ on } V. \end{cases} \tag{34}$$

The preimage $(-\rho_U \omega, \rho_V \omega)$ seems palusible, but $-\rho_U \omega$ is not defined on U since ω only has value on $U \cap V$, while we can extend $-\rho_V \omega$ on U by zero. Therefore it is easier to apply the pasting lemma to $-\rho_V \omega$ to find the preimage and it would be hard for us to find the preimage of $-\rho_U \omega$. □

Example 2. *Let us calculate $H^q(\mathbb{S}^1)$. $\mathbb{S}^1$ is covered by two open intervals U, V which look like "(" and ")". Their intersections are also two open intervals which look like $\cap$ and $\cup$. These intervals are all diffeomorphic to $\mathbb{R}$. Since the manifold is 1-dim, the only non-zero spaces are H^0 and H^1.*

$$0 \to H^0(\mathbb{S}^1) \xrightarrow{(i_U^*, i_V^*)} H^0(\mathbb{R}) \oplus H^0(\mathbb{R}) \xrightarrow{\partial_V^* - \partial_U^*} H^0(\mathbb{R} \coprod \mathbb{R}) \xrightarrow{d^*} H^1(\mathbb{S}^1) \cdots \tag{35}$$

Since $H^0(\mathbb{R}) = \mathbb{R}$ and $H^1(\mathbb{R}) = 0$, the exact sequence is

$$0 \to H^0(\mathbb{S}^1) \xrightarrow{f} \mathbb{R}^2 \xrightarrow{\partial_V^* - \partial_U^*} \mathbb{R}^2 \xrightarrow{g} H^1(\mathbb{S}^1) \to 0 \to 0 \tag{36}$$

Since f is injective,

$$H^0(\mathbb{S}^1) = \mathrm{im} f = \ker(\partial_V^* - \partial_U^*). \tag{37}$$

Since $(\partial_V^* - \partial_U^*)(\omega, \eta) = (\eta - \omega, \eta - \omega)$ on $U \cap V$, its matrix representation is just

$$\begin{bmatrix} -1 & 1 \\ -1 & 1 \end{bmatrix}. \tag{38}$$

The kernel is 1-dim and so isomphic to $\mathbb{R}$. Hence $H^0(\mathbb{S}^1) = \mathbb{R}$. Now since g is onto,

$$H^1(\mathbb{S}^1) = \mathrm{im} g = \mathbb{R}^2/\ker g = \mathbb{R}^2/\mathrm{im}(\partial_V^* - \partial_U^*) = \mathbb{R}. \tag{39}$$

Example 3. *Let us calculate the generator(s) of $H^1(\mathbb{S}^1)$. Since this is a $1-d$ vector space, this is equivalent to finding a non-zero element(vector) in $H^1(\mathbb{S}^1)$. To do this, let us consider α closed in $\Omega^0(U \cap V)$ such that $d^*\alpha$ is not exact. Since $\ker d^* = \mathrm{im}(\partial_1^* - \partial_0^*)$, we only need to find α that is not an image of $(\partial_1^* - \partial_0^*)$(some constant on $U \cap V$.) Suppose α is m on one segment "(" of $U \cap V$ and n on ")". Then*

$$d^*[\alpha] = d^*[m\text{“(”} + n\text{“)”}] = d^*[(m-n)\text{“(”}] + \cancel{d^*[n\text{“(”} + n\text{“)”}]} = (m-n)d^*[\text{“(”}]. \tag{40}$$

Therefore a generator is $\alpha = 1$ on "(" and 0 on ")". The image is just $-d\rho_V$ on "(" side of $U \cap V$.

Example 4. *Let us calculate H^q of the Möbius strip M. The Möbius strip has the structure of $\mathbb{R} \times \mathbb{R}$ identifying (t, v) with $(t + 2\pi, -v)$. In other words, it is*

$$\{(t, v) \in \mathbb{R} \times \mathbb{R} : (t, v) = (t + 2\pi, -v)\}. \tag{41}$$

This can be covered by two open strips $U \times \mathbb{R}$ and $V \times \mathbb{R}$ whose intersections are another two disjoint open strips $X \times \mathbb{R}$ and $Y \times \mathbb{R}$, under the condition that if $(x, \partial_U^* v) = (x, \partial_V^* v)$ then $(y, \partial_U^* v) = (y, -\partial_V^* v)$. The non-zero cohomologies are $H^{0,1,2}$, so

$$0 \to H^0(M) \xrightarrow{f} \mathbb{R}^2 \xrightarrow{\partial_V^* - \partial_U^*} \mathbb{R}^2 \xrightarrow{g} H^1(M) \to 0 \to 0 \to H^2(M) \to 0 \to 0 \tag{42}$$

The only difference between this and the case for $\mathbb{S}^1$ is the computation of the image(or kernel) of $\partial_V^* - \partial_U^*$. This maps (ω, η) to $(\partial_V^*\eta(x) - \partial_U^*\omega(x), \partial_V^*\eta(y) - \partial_U^*\omega(y))$. The image is $(0, 0)$ only if $(x, \partial_V^*\eta) = (x, \partial_U^*\omega)$, and $\partial_V^*\eta(y) - \partial_U^*\omega(y) = 2\partial_V^*\eta(y) = 0$. This would mean $\eta = 0$ on V and similarly $\omega = 0$ on U. So $(\omega, \eta) = (0, 0)$ so the kernel of $\partial_V^* - \partial_U^*$ is empty. This would mean $H^0(M) = 0$ and that $H^1(M) = \mathbb{R}^2/\mathrm{im}(\partial_V^* - \partial_U^*) = 0$. Obviously $H^2(M) = 0$. So $H^q(M) = 0$.

Definition 8. *Let $M = U \cup V$, U, V open. Then we have*

$$U \cap V \xrightarrow[\partial_V]{\partial_U} U \coprod V \xrightarrow{(i_U, i_V)} U \cup V = M \tag{43}$$

where ∂_j, i_j are inclusion maps. Applying the covariant functor Ω_c^*: $M \to$ vector space of compactly supported forms, and Mor(inclusions) $\to \Omega_c^*$(Mor(inclusions)) *then we have*

$$\Omega_c^*(M) \xleftarrow{\partial_*^U + \partial_*^V} \Omega_c^*(U) \oplus \Omega_c^*(V) \xleftarrow{(-i_*^U, i_*^V)} \Omega_c^*(U \cap V). \tag{44}$$

This is the Mayer-Vietoris sequence for a covariant functor. Here i_*^j, ∂_*^j are the extention maps which extend the form by 0.

Comment 3.

$$F^*\omega(x) = \omega \circ F(x); \quad F_*\omega(x) = F(\omega)(x). \tag{45}$$

The pullback pulls the domain of ω to the domain of F; The pushforward pushes the domain of ω to the domain of the image of F. For example, take $U \subset M$ and ω_M, ω_U defined on M, U. Let $i_U : U \hookrightarrow M$ be the inclusion map. $i_U^*\omega_M$ has non-changed value but the domain is shrunken to U; $i_*^U\omega_U$ has non-changed value on U but the domain is extended to M by 0. So when i_U is the inclusion map, i^* is the restriction and i_* is the extension by 0.

Contrarily, let $\pi_U : U \times V \to U$ be the projection map. Then π_U^* is the extension and π_*^U is the restriction.

Proposition 3. *The covariant M-V sequence is exact.*

Proof. To see injectiveness, let $\omega \in \Omega_c^*(U \cap V)$ and

$$(-i_*^U, i_*^V)\omega = (-\omega, \omega) = (0,0) \Rightarrow \omega = 0. \tag{46}$$

To see exactness, first $(\partial_*^U + \partial_*^V) \circ (-i_*^U, i_*^V)\omega = 0$ is obvious. Now let $0 = (\partial_*^U + \partial_*^V)(\omega, \eta) = \omega + \eta$. While ω is defined on U and η on V, to have a zero sum they must be zero on $U \setminus V$ and $V \setminus U$ respectively. In other words they are only nonzero on $U \cap V$. Moreover $\omega = -\eta$ on $U \cap V$. Then

$$(\omega, \eta) = (-i_*^U, i_*^V)(\eta|_{U \cap V}). \tag{47}$$

To see surjectiveness, let $\omega \in \Omega_c^*(M)$, it is direct to see $\omega = \rho_U \omega + \rho_V \omega$, and that $\rho_U \omega \in \Omega_c^*(U)$. □

Corollary 2. *By snake lemma, we have a long exact sequence*

$$\cdots H_c^{i+1}(U \cap V) \xleftarrow{d_*} H_c^i(M) \xleftarrow{\partial_*^U + \partial_*^V} H_c^i(U) \oplus H_c^i(V) \xleftarrow{(-i_*^U, i_*^V)} H_c^i(U \cap V) \cdots \tag{48}$$

Comment 4. *Since $\mathbb{S}^1$ is compact, $H_c^*(\mathbb{S}^1)$ should not differ from $H^*(\mathbb{S}^1)$.*

Proposition 4. *Let the projection and section functions be*

$$\begin{aligned} \pi : M \times \mathbb{R} \to M : &\quad (x,t) \mapsto x; \\ s_0 : M \to M \times \mathbb{R} : &\quad x \mapsto (x, 0). \end{aligned} \tag{49}$$

Then we have them induced:

$$\begin{aligned} \pi^* : \Omega^q(M) \to \Omega^q(M \times \mathbb{R}) : &\quad \omega(x) \mapsto \omega \circ \pi(x,t) = \omega(x); \\ s_0^* : \Omega^q(M \times \mathbb{R}) \to \Omega^q(M) : &\quad \alpha(x,t) + f(x,t)dt \wedge \eta(x) \mapsto \alpha(x, 0). \end{aligned} \tag{50}$$

$$\pi^* s_0^* = \mathrm{id}_{H^q(M \times \mathbb{R})}, \quad s_0^* \pi^* = \mathrm{id}_{H^q(M)}. \tag{51}$$

This will lead to the fact that

$$H^q(M) = H^q(M \times \mathbb{R}). \tag{52}$$

Proof. Obviously $s_0^* \pi^* = 1$. We would like to show that $\pi^* s_0^* = 1$. The obstruction is that, some information is lost in the process of $\pi^* s^*$ but none is lost in the process of $s_0^* \pi^*$. Let us focus on the $\{x_0\} \times \mathbb{R}$ space(fiber). The form on the fiber is mapped from $f(x_0, t)dt$ to $0dt$. (We write dt to indicate that it is in $\Omega^q(\{x_0\} \times \mathbb{R})$). And we would like to show that they are equivalent.

Lemma 1. $H^1(\mathbb{R}) = 0$.

Proof. Let $g(x)dx \in \Omega^1(\mathbb{R})$, it is automatically closed. Define

$$f(x) = \int_0^x g(u)du \Rightarrow df = g(x)dx. \tag{53}$$

Then any form on $\Omega(\mathbb{R})$ is exact. □

Therefore, the exact form between $f(x_0, t)dt$ and $0dt$ is simply

$$d\left(\int_0^t f(x_0, \tau)d\tau\right). \tag{54}$$

This leads to the following definition:

Definition 9. *The homotopy operator K is defined as*

$$\begin{aligned} K : \Omega^q(M \times \mathbb{R}) &\to \Omega^{q-1}(M \times \mathbb{R}), \\ \alpha(x,t) + f(x,t)dt \wedge \eta(x) &\mapsto \int_0^t f(x,\tau)d\tau \cdot \eta(x). \end{aligned} \tag{55}$$

Going back from the fiber $\{x_0\} \times \mathbb{R}$ to the whole space $M \times \mathbb{R}$, $\omega = \alpha(x,t) + f(x,t)dt \wedge \eta(x)$ and $\alpha(x,0)$ differ only by

$$d_t \int_0^t f(x,\tau)d\tau \wedge \eta(x) + [\alpha(x,t) - \alpha(x,0)] = d_t \int_0^t f(x,\tau)d\tau \wedge \eta(x) + \int_0^t d_\tau \alpha(x,\tau) \tag{56}$$

Now, let us look at

$$dK\omega = d_t \int_0^t f(x,\tau)d\tau \wedge \eta(x) + \left(d_M \int_0^t f(x,\tau)d\tau\right) \wedge \eta(x) + \int_0^t f(x,\tau)d\tau \cdot d\eta \tag{57}$$

$$= d_t \int_0^t f(x,\tau)d\tau \wedge \eta(x) + \int_0^t d\tau \wedge d_M f(x,\tau) \wedge \eta(x) + \int_0^t f(x,t)d\tau \cdot d\eta \tag{58}$$

However the last two terms are not what we want.

$$Kd\omega = \int_0^t d_\tau \alpha + \int_0^t d_M f(x,\tau) \wedge d\tau \wedge \eta(x) - \int_0^t f(x,\tau)d\tau \cdot d\eta \tag{59}$$

Hence

$$(dK + Kd)\omega = (1 - \pi^* s_0^*)\omega. \tag{60}$$

Then for a closed ω, we have $(1 - \pi^* s_0^*)\omega = dK\omega = 0$. □

Corollary 3. *Pullbacks of homotopic maps are equal in cohomology.*

Proof. Let the homotopy be $F(x,t)$.

$$F(x,1)^* = s_1^* F^* = (\pi^*)^{-1} F^* = s_0^* F^* = F(x,0)^*. \tag{61}$$

□

Corollary 4. *If M, N are homotopic: exist continuous $f : M \to N$ and $g : N \to M$ such that $f \circ g$ and $g \circ f$ are identity maps. Then $H^q(M) = H^q(N)$.*

Proof. $f^* g^* = 1_{H^q(M)}$ and $g^* f^* = 1_{H^q(N)}$ implies $g^* = (f^*)^{-1}$. These linear maps hence are isomorphisms. □

Theorem 2 (Poincaré lemma)**.** *For a contractible manifold M,*

$$H^q(M) = \begin{cases} \mathbb{R}, & q = 0; \\ \{0\}, & q > 0. \end{cases} \tag{62}$$

Theorem 3. *In $\mathbb{R}^n$ all closed forms are exact. So for $\omega = d\eta$ closed, η can be computed by*

$$\eta = -K_1 H^* \omega, \quad H(x,t) = tx. \tag{63}$$

Proof. Since $(1 - \pi^* s_1^*)\omega = (dK_1 + K_1 d)\omega = dK_1\omega$, so here K_1 performs the integral from 1 to t. Now,

$$\omega = 1^*\omega = H^*(x,1)\omega = s_1^* H^* \omega \Rightarrow \pi^*\omega = (1 - dK_1)H^*\omega \Rightarrow \cancel{s_0^* \pi^*} \omega = \cancel{s_0^* H^* \omega} - s_0^* dK_1 H^* \omega. \tag{64}$$

Hence

$$\omega = -s_0^* dK_1 H^* \omega = -ds_0^* K_1 H^* \omega \Rightarrow \eta = -s_0^* K_1 H^* \omega = \int_0^1 \omega(tx)dt. \tag{65}$$

□

Comment 5. *Some of these closed forms of compact support are d of non-compactly supported exact forms. Hence, $H_c^q(M)$ might be different.*

Proposition 5. *We define the pushforwards:*

$$\begin{aligned} \pi_* : \Omega^q(M \times \mathbb{R}) \to \Omega^{q-1}(M) : \quad & \alpha(x,t) + f(x,t)dt \wedge \eta(x) \mapsto \int_{\mathbb{R}} f(x,t)dt \cdot \eta; \\ e_* : \Omega^{q-1}(M) \to \Omega^q(M \times \mathbb{R}) : \quad & \eta(x) \mapsto \eta(x) \wedge e(t)dt, \quad \int_{\mathbb{R}} e = 1. \end{aligned} \tag{66}$$

Then $\pi_* e_*$ and $e_* \pi_*$ are identity maps in cohomology. This will result in

$$H_c^q(M \times \mathbb{R}) = H_c^{q-1}(M). \tag{67}$$

Proof. That π_*, e_* commute with d is easy to show and $\pi_* e_* = 1$ is obvious. First we look at the following lemma:

Lemma 2. *$H_c^1(\mathbb{R}) = \mathbb{R}$ and its generator is a $1-$bump function.*

Proof. We show that $\ker \int_{\mathbb{R}}$ is the space of all compactly supported exact forms:

$$\int_{\mathbb{R}} df = \int_{\mathbb{R}} \partial_x f dx = f(\infty) - f(-\infty) = 0. \tag{68}$$

Now if

$$\int_{\mathbb{R}} g(x)dx = 0, \tag{69}$$

let

$$f = \int_{-\infty}^{x} g(x)dx, \text{ whose compact support is } [-R, R], \tag{70}$$

then

$$df = g(x)dx, \quad f(-R) = f(R) = 0. \tag{71}$$

Hence, two forms are equivalent if their integral on $\mathbb{R}$ coincide. Then the forms are classified by their intgeral on $\mathbb{R}$, which is a real number, and a generator is $e(x)dx$. □

Comment 6. *The idea of the definition of e_* comes from the following: $f(x_0,t)dt$ is mapped to $B(x_0)$ by π_* where*

$$B(x_0) = \int_{\mathbb{R}} f(x_0,t)dt. \tag{72}$$

Now we would like to restore a form $g(x_0,t)dt$ equivalent to $f(x_0,t)dt$ from $B(x_0)$. We know that two forms are equivalent if their integral do not differ. A simple choice for gdt would be $Be(t)dt$, whose integral is B—the same as the integral of $f(x_0,t)dt$. Hence we define

$$e_* \eta(x_0) := \eta(x_0) e(t)dt. \tag{73}$$

Now we know that, on a fiber $\{x_0\} \times \mathbb{R}$ two forms ω, η of the same integral value differ by an exact form of compact support:

$$d\left(\int_{-\infty}^{t} \omega(x_0,t) - \eta(x_0,t)\right), \tag{74}$$

and hence on this fiber,

$$\int_{-\infty}^{t} f(x_0,\tau)d\tau \cdot \eta(x_0) - \int_{-\infty}^{t} B(x_0)e(\tau)d\tau \cdot \eta(x_0) \tag{75}$$

is exact. This leads to the following definition:

Definition 10. *The homotopy operator K is defined as*

$$\begin{aligned} K : \Omega_c^q(M \times \mathbb{R}) &\to \Omega_c^{q-1}(M \times \mathbb{R}) \\ \alpha(x,t) + f(x,t)dt \wedge \eta(x) &\mapsto \left(\int_{-\infty}^t f - B(x)e \right) \eta. \end{aligned} \tag{76}$$

Going back from the fiber to the whole space, $e_* \pi_* \omega = Be(t)dt \wedge \eta$ and $\omega = \alpha(x,t) + f dt \wedge \eta$ only differ by

$$d_t \left(\int_{-\infty}^t f(x,\tau)d\tau - B(x)e(\tau)d\tau \right) \wedge \eta + \int_{-\infty}^t d_\tau \alpha, \tag{77}$$

just like the previous case for the non-compactly supported case. In a more compact way, it writes

$$d_t \int_{-\infty}^t F(x,\tau)d\tau \wedge \eta + \int_{-\infty}^t d_\tau \alpha, \quad F(x,t) = f(x,t) - B(x)e(t). \tag{78}$$

$$d_M F = d_M f - e(t) d_M \int_{\mathbb{R}} f(x,t)dt = d_M f - e(t) \int_{\mathbb{R}} dt \wedge d_M f(x,t) \tag{79}$$

Now,

$$dK\omega = d \int_{-\infty}^t F d\tau \wedge \eta = d_t \int_{-\infty}^t F d\tau \wedge \eta + \int_{-\infty}^t d\tau \wedge d_M F \wedge \eta + \int_{-\infty}^t F d\tau \wedge d\eta. \tag{80}$$

And

$$d_M F(x,t) = d_M f - e(t) d_M B \tag{81}$$

$$\begin{aligned} K d\omega &= K(d_t \alpha + d_M f \wedge dt \wedge \eta - f dt \wedge d\eta) \\ &= \int_{-\infty}^t d_\tau \alpha + \int_{-\infty}^t d_M F \wedge d\tau \wedge \eta - \int_{-\infty}^t F d\tau \cdot d\eta. \end{aligned} \tag{82}$$

□

Corollary 5. *We know that e_* sends a non-exact form on $\Omega_c^{n-1}(\mathbb{R}^{n-1})$ to a non-exact form on $\Omega_c^n(\mathbb{R}^n)$ since it is bijective. Then by induction we have the generator for $\Omega_c^n(\mathbb{R}^n)$ being $a(x^1)a(x^2)\cdots a(x^n)dx^1 \cdots dx^n$.*

Theorem 4 (Poincaré Lemma for compactly supported forms)**.**

$$H_c^q(\mathbb{R}^n) = \begin{cases} \mathbb{R}, & q = n; \\ \{0\}, & \textit{otherwise}. \end{cases} \tag{83}$$

Comment 7. *We see that $H(M)$ and $H(M \times \mathbb{R})$ are related by an integration along the fiber. So $H(M \times \mathbb{R}^2)$ is related to $H(M)$ by integrating along $\mathbb{R} \times \{0\}$ first and then $\{0\} \times \mathbb{R}$. This is equivalent to just integrating along the whole fiber $\mathbb{R}^2$. So in fact $H(M)$ and $H(M \times \mathbb{R}^n)$ is simply related by an integration along the fiber map.*

Definition 11. *Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be proper, $f^* : H_c^n(\mathbb{R}^n) \to H_c^n(\mathbb{R}^n)$ and hence $[f^*\omega] = [r\omega]$, $\forall \omega$. This constant real number r is defined to be the degree of f. Alternatively, for a bump-n form α,*

$$\int_{\mathbb{R}^n} f^* \alpha = \int_{\mathbb{R}^n} r\alpha + d\eta = r. \tag{84}$$

Lemma 3. *The image of a proper map is closed.*

Proof. Let $f(x^n) \to y$. Then $\{y, f(x^1), f(x^2), ...\}$ is compact. The preimage K is hence compact and contains $x^1, x^2, ...$ By B-W theorem we have a convergent subsequence $X^1, X^2, ...$ converging to $X \in K$. Then $f(x^n) \to f(X)$ and so $f(X) = y$. □

Proposition 6. *If f is not surjective, it has degree 0.*

Proof. Suppose $p \notin f(\mathbb{R}^n)$ and since $f(\mathbb{R}^n)$ is closed, some open $U \cap f(\mathbb{R}^n) = \emptyset$, then let $\alpha = \rho_U dx^I$ then the integral is 0. □

Proposition 7. *The degree of a map is an integer.*

Proof. Let f be surjective, or the degree is 0. (By Sard's Theorem) We can find a regular value $q \in \mathbb{R}^n$ of f. Let $f(p_i) = q$ and then $df_{p_i} : T_{p_i}M \to T_qM$ is surjective and hence bijective. By inverse function theorem, there exists an open neighbourhood U_i of p which is diffeomorphic to $V \ni q$ with the diffeomorphism being f locally.

Now, $f^{-1}(\{q\})$ is a discrete set so it contains $p_1, ..., p_n$ and we obtain $U_1, ..., U_n$ diffeomorphic to V. Choose α to be supported in V, then

$$\int_{\mathbb{R}^n} f^*\alpha = \sum_i \int_{U_i} f^*\alpha = \sum_i \operatorname{sgn}(\det(df_{p_i})) \cancel{\int_{U_i}} \alpha. \tag{85}$$

□

Comment 8. *The key is to make f a diffeomorphism such that it can be taken out of the integral. Therefore, this proof can be summarized as: Because f can be locally a diffeomorphism, f^* turns into an integer.*

Definition 12. *An open cover U_i of a manifold M is called a good cover if any finite intersection is diffeomorphic to $\mathbb{R}^n$. M is said to be of finite type if it has a finite good cover.*

Proposition 8. *Any Riemannian manifold has a good cover.*

Proof. $\exp_p$ is a local diffeomorphism between $\mathcal{B}_r(\mathbf{0}) \subset T_pM \simeq \mathbb{R}^n$ and some $U \subset M$. Let $E(v^ie_i) := (v^1, ..., v^n)$ where e_i form an orthonormal basis in T_pM. Then $\phi_p := E \circ \exp_p^{-1}$ is a chart map. Then for $v = v^ie_i$ and then $\exp_p(v) \in M$,

$$\phi_p \exp_p(v) = (v^1, ..., v^n) \in \mathbb{R}^n. \tag{86}$$

Then we have

$$\phi_p \exp_p(\mathcal{B}_r(\mathbf{0})) = B_r(\mathbf{0}) \subset \mathbf{R}^n. \tag{87}$$

In other words, if the chart maps any point in M to some point in $B_r(\mathbf{0}) \subset \mathbf{R}^n$, it is in $\mathcal{B}_r(\mathbf{0})$. □

Definition 13. *A non-degenerate pairing $\langle \cdot, \cdot \rangle$ is a bilinear map : $V \otimes W \to \mathbb{R}$, which satisfies the non-degeneracy:*

$$\begin{aligned} \forall w \in W, \langle v, w \rangle = 0 \Leftrightarrow v = 0; \\ \forall v \in V, \langle v, w \rangle = 0 \Leftrightarrow w = 0. \end{aligned} \tag{88}$$

Proposition 9. *If we have a non-degenerate pairing between V and W, then $v \mapsto \langle v, \cdot \rangle$ defines an injective mapping from V to W^*, and alternatively, $w \mapsto \langle \cdot, w \rangle$ defines an injective mapping from W to V^*. The proof just follows from the non-degeneracy.*

Corollary 6. *If we have a non-degenerate pairing between V and W, $V \simeq W$.*

Lemma 4 (Five Lemma)**.** *Suppose we have the following commutataive diagram between abelian groups(vector spaces) and homomorphisms(linear maps), with each row being exact. Then if $\alpha, \beta, \delta, \epsilon$ are isomorphisms, we have γ being an isomorphism.*

$$\begin{tikzcd} \arrow[r] & A \arrow[r] \arrow[d, "\alpha"] & B \arrow[r] \arrow[d, "\beta"] & C \arrow[r] \arrow[d, "\gamma"] & D \arrow[r] \arrow[d, "\delta"] & E \arrow[r] \arrow[d, "\epsilon"] & \\ \arrow[r] & A' \arrow[r] & B' \arrow[r] & C' \arrow[r] & D' \arrow[r] & E' \arrow[r] & \end{tikzcd} \tag{89}$$

Proof. Let $\gamma(c) = 0 \in C'$. Then $f'_{BC}(b') = 0$ and due to exactness, we have $f'_{AB}(a') = b'$. Then for $\alpha(a) = a'$,

$$c = f_{BC} f_{AB}(a) = 0. \tag{90}$$

Then γ is injective. Now for $c' \in C'$, $f'_{DE} f'_{CD}(c') = 0$. So
$\delta^{-1} f'_{CD}(c') \in \ker f_{DE}$ due to exactness.
Also due to exactness $f'_{CD}(c') = \delta f_{CD}(c)$ for some c.
Due to commutativity, $\delta f_{CD}(c) = f'_{CD}\gamma(c)$.
Then $f'_{CD}(\gamma(c) - c') = 0$ so due to exactness $f'_{BC}(b') = \gamma(c) - c'$ for some b'.
Due to commutativity, $f'_{BC}(b') = \gamma f_{BC}\beta^{-1}(b')$.
Now we would have that

$$c' = \gamma(c) - \gamma f_{BC}\beta^{-1}(b') = \gamma[c - f_{BC}\beta^{-1}(b')]. \tag{91}$$

Then γ is surjective. □

Lemma 5. *For an orientable manifold, we can form a pairing between the following two M-V sequences:*

$$\begin{array}{ccccccc} \cdots H^i(M) & \xrightarrow{(i_U^*, i_V^*)} & H^i(U) \oplus H^i(V) & \xrightarrow{\partial_V^* - \partial_U^*} & H^i(U \cap V) & \xrightarrow{d^*} & H^{i+1}(M) \cdots \\ \otimes & & \otimes & & \otimes & & \otimes \\ \cdots H_c^{n-i}(M) & \xleftarrow{\partial_*^U + \partial_*^V} & H_c^{n-i}(U) \oplus H_c^{n-i}(V) & \xleftarrow{(-i_*^U, i_*^V)} & H_c^{n-i}(U \cap V) & \xleftarrow{d_*} & H_c^{n-i-1}(M) \cdots \end{array} \tag{92}$$

where the pairing between each two spaces $H^i(N)$ and $H_c^{n-i}(N)$ in a column is given by

$$\langle \omega, \eta \rangle := \int_N \omega \wedge \eta. \tag{93}$$

This lemma asserts that, this diagram commutes mod sign. The well-defined-ness for this pairing boils down to the Stokes' Theorem, which requires M to be orientable.

Proof. Here we have three blocks in total. We prove them separately. Let $\omega \in H^i(M)$ and $(\eta_U, \eta_V) \in H_c^{n-i}(U) \oplus H_c^{n-i}(V)$. The equality that

$$\int_U i_0^* \omega \wedge \eta_U + \int_V i_1^* \omega \wedge \eta_V = \int_M \omega \wedge (\partial_*^U \eta_U + \partial_*^V \eta_V) \tag{94}$$

is obvious by restricting the integral domain to the support. Let $(\omega_U, \omega_V) \in H^i(U) \oplus H_c^i(V)$ and $\eta \in H^i(U \cap V)$. The equality that

$$\int_{U \cap V} \partial_1^* \omega_V \wedge \eta - \partial_0^* \omega_U \wedge \eta = -\int_U \omega_U \wedge i_*^U \eta + \int_V \omega_V \wedge i_*^V \eta \tag{95}$$

is still obvious by restricting the integral domain to the support, which is $U \cap V$. Now let $\omega \in H^i(U \cap V)$, $\eta \in H^{n-i-1}(M)$. LHS:

$$\int_M d^* \omega \wedge \eta = \int_M -d\rho_V \wedge \omega \wedge \eta. \tag{96}$$

RHS:

$$\int_{U \cap V} \omega \wedge d_* \eta \tag{97}$$

and $d_*\eta$ is obtained from: take the preimage of η: $(\rho_U \eta, \rho_V \eta)$ then $d_*\eta$ is such that $(-i_*^U, i_*^V) d^* \eta = d(\rho_U \eta, \rho_V \eta)$. Hence $(-i_*^U d_* \eta, i_*^V d_* \eta) = (d\rho_U \wedge \eta, d\rho_V \wedge \eta)$. On $U \cap V$ we have $i_*^V d_* \eta = d_* \eta$, so the RHS is

$$\int_{U \cap V} \omega \wedge d\rho_V \wedge \eta = (-1)^{|\omega|} \int_{U \cap V} d\rho_V \wedge \omega \wedge \eta = (-1)^{|\omega|+1} \int_M -d\rho_V \wedge \omega \wedge \eta. \tag{98}$$

So LHS $= (-1)^*$ RHS. □

Theorem 5 (Poincaré Duality)**.** *If M is orientable and of finite type, there exists a non-degenerate pairing between $H^q(M)$ and $H_c^{n-q}(M)$. Hence*

$$H^q(M) \simeq H_c^{n-q}(M)^*. \tag{99}$$

Proof. We do induction on the cardinality of the finite good cover:
Base case: $M \simeq \mathbb{R}^n$. Then $H^q(M) = \mathbb{R}^{\delta_{0,q}}$ and $H_c^{n-q}(M)^* = \mathbb{R}^{\delta_{0,n-(n-q)}} = \mathbb{R}^{\delta_{0,q}}$.
Suppose Poincaré duality holds for any $p-$good cover.
Now, let

$$U_0 \cup ... \cup U_p = (U_0 \cup ... \cup U_{p-1}) \cup U_p = U \cup V = M$$

be a $p+1-$good cover. Then

$$\{U_0 \cup ... \cup U_{p-1}\} \cap U_p = \{U_0 \cap U_p, U_1 \cap U_p, ..., U_{p-1} \cap U_p\} = U \cap V$$

forms a manifold with $p-$good cover. Since M is orientable we have

$$\begin{array}{ccccc} \to H^{i-1}(U) \oplus H^{i-1}(V) & \to H^{i-1}(U \cap V) & \to H^i(M) & \to H^i(U) \oplus H^i(V) & \to H^i(U \cap V) \to \\ \updownarrow & \updownarrow & \otimes & \updownarrow & \updownarrow \\ \leftarrow H_c^{n-i+1}(U) \oplus H_c^{n-i+1}(V) & \leftarrow H_c^{n-i+1}(U \cap V) & \leftarrow H_c^{n-i}(M) & \leftarrow H_c^{n-i}(U) \oplus H_c^{n-i}(V) & \leftarrow H_c^{n-i}(U \cap V) \leftarrow \end{array} \tag{100}$$

commutes (mod sign) and four pairings nondegenerate. Since

$$\leftarrow H_c^{n-i+1}(U) \oplus H_c^{n-i+1}(V) \leftarrow H_c^{n-i+1}(U \cap V) \leftarrow H_c^{n-i}(M) \leftarrow H_c^{n-i}(U) \oplus H_c^{n-i}(V) \leftarrow H_c^{n-i}(U \cap V) \leftarrow \tag{101}$$

being exact leads to

$$\to H_c^{n-i+1}(U)^* \oplus H_c^{n-i+1}(V)^* \to H_c^{n-i+1}(U \cap V)^* \to H_c^{n-i}(M)^* \to H_c^{n-i}(U)^* \oplus H_c^{n-i}(V)^* \to H_c^{n-i}(U \cap V)^* \to \tag{102}$$

being exact with the maps taken adjoint, we have

$$\begin{array}{ccccc} \to H^{i-1}(U) \oplus H^{i-1}(V) & \to H^{i-1}(U \cap V) & \to H^i(M) & \to H^i(U) \oplus H^i(V) & \to H^i(U \cap V) \to \\ \updownarrow & \updownarrow & \downarrow & \updownarrow & \updownarrow \\ \to H_c^{n-i+1}(U)^* \oplus H_c^{n-i+1}(V)^* & \to H_c^{n-i+1}(U \cap V)^* & \to H_c^{n-i}(M)^* & \to H_c^{n-i}(U)^* \oplus H_c^{n-i}(V)^* & \to H_c^{n-i}(U \cap V)^* \to \end{array} \tag{103}$$

commutes (mod sign) and four isomorphisms. By five lemma the middle map is an isomorphism, since five lemma in fact allows mod sign in the commutativity, as the kernel is not affected if the map is multiplied by -1. □

Comment 9. *The pairing is defined on orientable manifolds and the proof uses the induction on the finite good cover. These are where the assumptions lie.*

Definition 14. *The degree of a proper map between compact spaces $f : M^n \to N^n$ is defined as*

$$\int_N f^*\alpha \tag{104}$$

for some generator α of $H^n(N)$. Since $\mathbb{R} = H^0(M) = H_c^n(M)$, the map has fixed value in $\mathbb{R}$ and following similar arguments as before, this would be an integer.

Definition 15. *Let S^k closed $\subset M^n$ where M is oriented with finite good cover. The Poincaré dual of S in M is defined as $[\eta_S]$, where*

$$\int_S i_S^*\omega = \int_M \omega \wedge \eta_S \text{ for } \omega \text{ with compact support.} \tag{105}$$

The existence of Poincaré dual is guaranteed since $H^{n-k}(M) = [H^k(M)_c]^*$.

$$\int_S i_S^*\omega \text{ is a linear functional on } M \Longrightarrow \int_S i_S^*(\cdot) \in [H_c^k(M)]^*. \tag{106}$$

So there exists $\eta_S \in H^{n-k}(M)$ such that

$$\int_S i_S^*(\cdot) = \langle \eta_S, \cdot \rangle = \int_M (\cdot) \wedge \eta_S. \tag{107}$$

If ω is not with compact support it may not be integrable over S.

Definition 16. *If S is now compact, ω does not need to be compact. So that we may consider any form ω. Then*

$$\int_S i_S^*(\cdot) \in [H^k(M)]^* = H_c^{n-k}(M). \tag{108}$$

So $[\eta_S]$ is with compact support. This η_S is denoted by η'_S and is called the closed Poincaré dual.

Comment 10. *In fact, this is similar to the Riestz-representation theorem in linear algebra. For any linear functional $L(v), v \in V$, there is a $u \in V$ such that*

$$F(v) = \langle u, v \rangle . \tag{109}$$

Here the restriction is more loose, since the pairing is between different vector spaces.

Theorem 6 (Kunneth Formula)**.** *Let M has a finite good cover, then for F a manifold, we have*

$$H^q(M \times F) = \bigoplus_{i+j=q} H^i(M) \otimes H^j(F). \tag{110}$$

Proof. We construct an isomorphism between the two spaces. Let the projection maps be $\pi : M \times F \to M$ and $\rho : M \times F \to F$. It is natural to have

$$\psi^* : \bigoplus_{i+j=q} H^i(M) \otimes H^j(F) \to H^q(M), \quad \omega \otimes \eta \mapsto \pi^*\omega \wedge \rho^*\eta. \tag{111}$$

First we consider the following M-V sequence:

$$\begin{aligned} \bigoplus_{i=1}^n H^i(U \cup V) \otimes H^{q-i}(F) &\xrightarrow{(i_U^*, i_V^*)} \bigoplus_{i=1}^n H^i(U) \otimes H^{q-i}(F) \oplus H^i(V) \otimes H^{q-i}(F) \\ &\xrightarrow{\partial_V^* - \partial_U^*} \bigoplus_{i=1}^n H^i(U \cap V) \otimes H^{q-i}(F) \xrightarrow{d_M^*} \bigoplus_{i=1}^n H^{i+1}(U \cup V) \otimes H^{q-i-1}(F) \end{aligned} \tag{112}$$

Connected vertically by ψ^* with

$$H^q((U \cup V) \times F) \xrightarrow{(i_{U\times F}^*, i_{V\times F}^*)} H^q(U \times F) \oplus H^q(V \times F) \xrightarrow{\partial_{V\times F}^* - \partial_{U\times F}^*} H^q((U \cap V) \times F) \xrightarrow{d_{M\times F}^*} H^q((U \cup V) \times F) \tag{113}$$

We show that this diagram commutes. First we consider ω defined on $U \cup V$.

$$\begin{aligned} \psi^*(i_U^*, i_V^*)(\omega \otimes \eta) &= \psi^*(\omega_U \otimes \eta, \omega_V \otimes \eta) \\ &= (\pi_U^*\omega_U \wedge \rho^*\eta, \pi_V^*\omega_V \wedge \rho^*\eta); \\ (i_{U\times F}^*, i_{V\times F}^*)\psi^*\omega &= (i_{U\times F}^*, i_{V\times F}^*)(\pi_{U\cup V}^*\omega \wedge \rho^*\eta) \\ &= (\pi_U^*\omega_U \wedge \rho^*\eta, \pi_V^*\omega_V \wedge \rho^*\eta) \end{aligned} \tag{114}$$

since $\pi_{U\cup V}^*$ is the extension of domain and $i_{U\times F}^*$ is the restriction to the strip.

Next we consider ω^U on U and ω^V on V. Then

$$\begin{aligned} \psi^*(\partial_V^* - \partial_U^*)(\omega^U, \omega^V) &= \pi_{U\cap V}^*(\partial_V^*\omega^V - \partial_U^*\omega^U) \wedge \rho^*\eta \\ &= \pi_{U\cap V}^*(\omega_{U\cap V}^V - \omega_{U\cap V}^U) \wedge \rho^*\eta; \\ (\partial_{V\times F}^* - \partial_{U\times F}^*)\psi^*(\omega_U, \omega_V) &= (\partial_{V\times F}^* - \partial_{U\times F}^*)(\pi_U^*\omega^U, \pi_V^*\omega^V) \wedge \rho^*\eta \\ &= (\pi_{U\cap V}^*\omega_{U\cap V}^V - \pi_{U\cap V}^*\omega_{U\cap V}^U) \wedge \rho^*\eta. \end{aligned} \tag{115}$$

since π_U^* is the extension of domain to fibers on U and $\partial_{U\times F}^*$ is the restriction to an intersection fiber.

At last we consider ω on $U \cap V$.

$$\begin{aligned} \psi^* d_M^* \omega &= \pi_{U\cup V}^*(dn_V \wedge i_*^{U\cup V}\omega) \wedge \rho^*\eta \\ &= dn_{(U\cup V)\times F} \wedge (\pi_{U\cup V}^* i_*^{U\cup V}\omega) \wedge \rho^*\eta; \\ d_{M\times F}^*\psi^*\omega &= dn_{(U\cup V)\times F} \wedge i_*^{(U\cap V)\times F}\pi_{U\cap V}^*\omega \wedge \rho^*\eta, \end{aligned} \tag{116}$$

since we can extend by 0 on M and extend the fiber or extend the fiber and then extend by 0 on $M \times F$.

Now we do induction on the number of good covers that the isomorphism holds. The base case is when $M = \mathbb{R}^n$, then $H^q(\mathbb{R}^n \times F) = H^q(F) = \mathbb{R}^{\delta_{i,0}} \otimes H^{q-i}(F)$. The following is exactly the same as the proof in Poincaré duality by using Five lemma. □

Proposition 10. *Let M has finite good cover and F a manifold,*

$$H_c^q(M \times F) = \bigoplus_{i+j=q} H_c^i(M) \otimes H_c^j(F). \tag{117}$$

Proof.

$$\psi_* : \bigoplus_{i+j=q} H_c^i(M) \otimes H_c^j(F) \to H_c^q(M \times F), \quad \omega \otimes \eta \mapsto i_*^M\omega \wedge i_*^F\eta. \tag{118}$$

Apply the same argument using the covariant M-V sequence. □

Definition 17. *Suppose we have $\pi : E \to M$ a vector bundle. For $f : N \to M$ we define the pullback bundle $f^{-1}(E)$ over N to be :*

$$\{(n, e) \in N \times E : \pi(e) = f(n)\}. \tag{119}$$

In other words, take any point $n \in N$, map this to $f(n) \in M$, on this point there is a fiber $\pi^{-1}(f(n))$, now grow this on N. This fiber bundle on N is $f^{-1}(N)$.

Theorem 7. *Let f_0 and f_1 be homotopic maps from Y to X, then*

$$f_0^{-1}E = f_1^{-1}E. \tag{120}$$

Proof. We know that an isomorphism between E_0 and E_1 over M is just a section of the fiber bundle $\mathrm{Iso}(E_0, E_1)$ over M. (A section is continuous.) The fibers are just $\mathrm{GL}(n, \mathbb{R})$.

First we take a times slice of $F_t^{-1}E$, which is a fiber bundle over $Y \times \{t\}$. So it is isomorphic to a fiber bundle E' over Y. Let π be the projection operator onto the space. Then we have

$$F^{-1}E = \pi^{-1}E' \quad \text{on the time slice } Y \times \{t\}. \tag{121}$$

In other words, there is a section on $\mathrm{Iso}(E_0, E_1)$ over $Y \times \{t\}$ which is also a section on $\mathrm{Hom}(E_0, E_1)$ over $Y \times \{t\}$. Now we choose a finite number (thanks to the compactness of Y) of trivializing open sets, then $\mathrm{Hom}(E_0, E_1)$ are trivialized to $\mathbb{R}^{n\times n}$. Then the trivialization map also sends the section on $\mathrm{Iso}(E_0, E_1)$ to a section on $\mathbb{R}^{n\times n}$ growing on this trivializing open set. Such a section has a continuous non-zero determinant. For t' close enough to t, the corresponding section must has a determinant close to the one at t such that it is also non-vanishing. Due to the compactness of $[0, 1]$, this argument is extended over the whole $[0, 1]$ line. □

Comment 11. *A homopoty is a continuous map $F : Y \times I \to X$, so $F^{-1}(E)$ is just a fiber bundle over $Y \times \mathbb{R}$. As long as two slices are close in time, they are isomorphic by the continuity of F. Since the timeline is compact, this relation is extended over the entire timeline. In the above proof, the rigorous part is just to trivialize the bundles such that the* Iso *section are represented by determinants—making the proof easier.*

Corollary 7. *Any fiber on a contractible manifold is trivial.*

Example 5. *Calculate* $\mathrm{Vect}_k(\mathbb{S}^1)$. *This is just* $[0,1]\times\mathbb{R}^k/f$, *where the equivalence relation is* $(0,v)\sim(1,f(v))$ *and* f *isomorphic. Now we show that* $/f$ *is the same as* $/g$ *iff* f *and* g *are homotopic in* $\mathrm{GL}(k,\mathbb{R})$. *Suppose* f,g *are homotopic through* $F_t\in\mathrm{GL}(n,\mathbb{R})$. *We show that the bundles are isomorphic through a map*

$$[t,v]_f\mapsto[t,F_t(v)]_g. \tag{122}$$

It is obvious that this is an isomorphism. We need to check the well-defined-ness at the endpoint.

$$\begin{aligned}[0,v]_f&\mapsto[0,F_0(v)]_g=[0,f(v)]_g=[1,gf(v)]_g\\ [1,f(v)]_f&\mapsto[1,gf(v)]_g\,.\end{aligned} \tag{123}$$

So this map is well-defined. Since all maps having the same sign of determinant are homotopic, there can only be two equivalence classes. Now suppose $/f$ *and* $/g$ *are the same. The isomorphism is fiberwise so* $[t,v]_f\mapsto[t,F_t(v)]_g$. *Following the above argument, we see that* $F_0=f$ *and* $F_1=g$ *so* f *and* g *are homotopic.*

Definition 18. *For a vector bundle* E *over* M, *the cohomology with vertical support* $H^q_{cv}(E)$ *is such that the forms are compactly supported fiberwise.*

Definition 19. *The integration along the fiber map* $\pi_*:H^q_{cv}(E)\to H^q(M)$ *is defined as*

$$\pi_*(\pi^*\omega\wedge f(\vec{t})d\vec{t})=\omega\int_{\mathbb{R}^n}f(\vec{t})d\vec{t}. \tag{124}$$

It is easy to check that the value is independent of the localization. This map is continuous if E *is oriented.*

Proposition 11. π_* *commutes with* d.

Proof. In the previous proof of Poincaré lemma, the integration along $\mathbb{R}$ fiber map commutes with d. Here it is almost the same. □

Proposition 12. *Let* E *be oriented and* τ *on* M, ω *on* E *then we have*

$$\pi_*(\tau\wedge\omega)=\tau\wedge\pi_*\omega. \tag{125}$$

Proof. This is obvious since π_* does nothing to τ. □

Lemma 6. *For* M *oriented,*

$$H^q_{cv}(M\times\mathbb{R}^n)=H^{q-n}(M). \tag{126}$$

Proof. This is almost the same as the proof of Poincaré lemma for compact support. □

Theorem 8 (Thom Isomorphism)**.** *For* E *oriented with finite good cover, we have*

$$H^q_{cv}(E)=H^{q-n}(M). \tag{127}$$

Proof. The proof again follows by induction on the cardinality of good covers and then appying the five lemma and the M-V sequence. What is left to show is just the commutative diagram:

$$\begin{tikzcd}
\cdots\to H^q_{cv}\left(E|_{U\cup V}\right)\arrow[d, phantom, "\Big\downarrow\pi_*"] \to H^q_{cv}\left(E|_U\right)\oplus H^q_{cv}\left(E|_V\right)\arrow[d, phantom, "\Big\downarrow\pi_*"] \to H^*_{cv}\left(E|_{U\cap V}\right)\arrow[d, phantom, "\Big\downarrow\pi_*"] \xrightarrow{d^*} H^{q+1}\left(E|_{U\cup V}\right)\arrow[d, phantom, "\Big\downarrow\pi_*"] \to\cdots\\
\cdots\to H^{q-n}(U\cup V)\to H^{q-n}(U)\oplus H^{q-n}(V)\to H^{q-n}(U\cap V)\xrightarrow[d^*]{} H^{q+1-n}(U\cup V)\to\cdots
\end{tikzcd} \tag{128}$$

The blocks involving restriction mappings are obvious since integration is linear. For the block involving d^*, we have

$$\pi_*d^*\omega=\pi_*(d\rho_U\wedge\omega)=d\rho_U\wedge\pi_*\omega=d^*\pi_*\omega. \tag{129}$$

□

Definition 20. *The Thom class Φ is an n-form on E such that $\pi_*\Phi = 1$. The existence comes from the orientability of E, where we have a non-vanishing $n+k$-form. Normalizing w.r.t. π_* and project it onto the fibers, and we would have Φ.*

Comment 12. *Now we may find the inverse of the Thom isomorphism π_*, since for ω closed on M,*

$$\pi_*(\omega \wedge \Phi) = \omega \cdot \pi_*\Phi = \omega. \tag{130}$$

Proposition 13. *Let E_1, E_2 be oriented over M, then*

$$\Phi(E_1 \oplus E_2) = \pi_1^*\Phi(E_1) \wedge \pi_2^*\Phi(E_2). \tag{131}$$

Proof. The Thom isomorphism π_* on $E_1 \oplus E_2$ is simply integrating along the two fibers consecutively. It is direct to see that

$$\pi_*^1\pi_*^2[\pi_1^*\Phi(E_1) \wedge \pi_2^*\Phi(E_2)] = \pi_*^1\pi_1^*\Phi(E_1) \wedge \pi_*^2\pi_2^*\Phi(E_2) = 1. \tag{132}$$

□

Definition 21. *Given $S^k \subset M^n$, the tubular neighborhood of S, denoted by $T_S \subset M$ is an open neighborhood of S that is diffeomorphic to a vector bundle of rank $n-k$ with S diffeomorphic to the zero section.*

Definition 22. *Given $S^k \subset M^n$, the normal bundle of S is denoted by NS, where$(TS)^\perp \subset TM$ grows on S.*

Lemma 7. *$NS = T_S$ is connected by an isomorphism f.*

Comment 13. *In fact, f just tilts the fibers growing on S.*

Theorem 9. *Consider*

$$H^q(S) \xrightarrow{\pi^*(\cdot)\wedge\Phi} H_{cv}^{q+n-k}(NS) \xrightarrow{f^*} H_{cv}^{q+n-k}(T_S) \xrightarrow{i_*^T} H_{cv}^{q+n-k}(M). \tag{133}$$

We have the Poincaré dual of S

$$\eta_S = i_*^T f^*\Phi. \tag{134}$$

Proof. Let $i_{S,T} : S \hookrightarrow T_S$ and $\pi_{T,S} : T_S \to S$. Let $\omega \in \Omega_c^k(M)$, then

$$\begin{aligned}\int_S i_S^*\omega &= \int_{NS} \pi^* i_S^*\omega \wedge \Phi = \int_{T_S} f^*\pi^* i_S^*\omega \wedge f^*\Phi \\ &= \int_M i_*^T(f^*\pi^* i_S^*\omega \wedge f^*\Phi) = \int_M f^*\pi^* i_S^*\omega \wedge i_*^T f^*\Phi.\end{aligned} \tag{135}$$

Now we show that $f^*\pi^* i_S^*$ is identity on T. Since $(i_S \circ \pi \circ f)$ is just the projection from T_S to S, it is homotopic to identity. So the above integral is reduced to

$$\int_M \omega \wedge i_*^T f^*\Phi + \int_M d\eta \wedge i_*^T f^*\Phi. \tag{136}$$

The latter one is zero, for it equals

$$\int_S i_S^*(d\eta) = \int_S d(i_S^*\eta) = 0. \tag{137}$$

by reversing our proof. Else we may verify this by direct computation. □

Comment 14 (Localization Principle)**.** *The support of η_S can be any tubular neighborhood of S. This is because $NS = T_S$ for any T_S.*

Definition 23. *$S, R \subset M$ intersect transversally if $T_xS \oplus T_xR = T_xM$ for any $x \in S \cap R$.*

Lemma 8. *$NS \oplus NR = N(S \cap R)$ if S, R intersect transversally.*

Proof. By Grassmann's identity,

$$\dim(M+N)+\dim(M\cap N)=\dim(M)+\dim(N). \tag{138}$$

This is obtained by taking a basis of $M\cap N$ and then extending it to M and N and $M+N$ respectively. So $n+\dim(T(S\cap R))=s+r\Rightarrow\dim(T(S\cap R))=s+r-n$ and so

$$\dim N(S\cap R)=n-(s+r-n)=(n-s)+(n-r)=\dim NS+\dim NR. \tag{139}$$

□

Proposition 14.

$$\eta_{S\cap R}=\eta_S\wedge\eta_R. \tag{140}$$

Proof.

$$\begin{aligned} i_*^T f^*\Phi(N(S\cap R)) &= i_*^T f^*\Phi(NS+NR) \\ &= i_*^T f^*(\Phi(NS)\wedge\Phi(NR)). \end{aligned} \tag{141}$$

□

Definition 24. *A smooth $F:N\to M$ is transversal to $S\subset M$ if $\forall x\in F^{-1}(S)$, we have*

$$dF(T_xN)+T_{F(x)}S=T_{F(x)}M. \tag{142}$$

Proposition 15. *Let F be orientation-preserving, and F is transversal to S and (of course) some T_S. Then $F^{-1}(T_S)$ is some tubular neighborhood of $F^{-1}(S)$, and*

$$\eta_{F^{-1}(S)}=F^*\eta_S. \tag{143}$$

Proof. For simplicity we pick Φ to be the generator along the fiber of T, instead of N. The fiber of T means the preimage of the fiber of the bundle N.

In fact, we need to shrink the tube such that the tube is onto for any S that is mapped to. This is how we are able to perform integration along the fibers. The orientation-preserving condition is to ensure the integral is 1 instead of -1. Now it is enough to show

$$\Phi(F^{-1}T)=F^*\Phi(T). \tag{144}$$

Take $F(x)\in S$, then consider T_x, the fiber at x.

$$1=\int_{T_x}\Phi(T)=\int_{F^{-1}(T_x)}F^*\Phi(T). \tag{145}$$

Hence we have

$$\eta_{F^{-1}(S)}=i_*^{F^{-1}(T)}F^*\Phi(T)=F^*(i_*^T\Phi(T))=F^*\eta_S. \tag{146}$$

□

Definition 25. *The volume form ω for $\Omega^n(M^n)$ is defined as such that*

$$\omega(e_1\wedge\cdots\wedge e_n)=1. \tag{147}$$

Proposition 16. *The volume form locally can be expressed by*

$$\omega=\sqrt{|g|}dx^1\wedge\cdots\wedge dx^n. \tag{148}$$

Proof. Let $g=QDQ^T$, then $a_ie_i=Q\partial_i$ and $a_1...a_n=\sqrt{|g|}$.

$$\begin{aligned} &\sqrt{|g|}dx^1\wedge\cdots\wedge dx^n(e_1\wedge\cdots\wedge e_n) \\ &\qquad=\sqrt{|g|}dx^1\wedge\cdots\wedge dx^n(Q\partial_1\wedge\cdots\wedge Q\partial_n)\frac{1}{a_1...a_n} \\ &\qquad=\sqrt{|g|}|Q|\frac{1}{\sqrt{|g|}}=1. \end{aligned} \tag{149}$$

□

Definition 26. *The Hodge star operator* $\star : \Lambda^i(V) \to \Lambda^{n-i}(V)$ *is defined as such that*

$$\alpha \wedge \star\beta = \langle \alpha, \beta \rangle \, \omega, \tag{150}$$

where ω *is the volume form.*

Definition 27. *The induced metric* $\hat{g}$ *on the cotangent space* T^*M *is defined as*

$$\langle \omega, \eta \rangle := \left\langle \omega^\sharp, \eta^\sharp \right\rangle_g . \tag{151}$$

Locally,

$$(\omega^\sharp)^i = g^{ij}\omega_j; \quad [g]^{ij} = ([g]^{-1})_{ij} \tag{152}$$

and

$$\langle \omega, \eta \rangle = \left\langle \omega^\sharp, \eta^\sharp \right\rangle_g = g_{ij} g^{i\mu}\omega_\mu g^{j\nu}\eta_\nu = \delta_i^\nu g^{i\mu}\omega_\mu\eta_\nu = g^{\mu\nu}\omega_\mu\omega_\nu. \tag{153}$$

Below our Hodge star will operate on $\Lambda(T^*M)$.

Proposition 17. $\star\star\omega = (-1)^{k(n-k)}\omega$.

Proof.

$$\langle \alpha, \beta \rangle = \langle \star\alpha, \star\beta \rangle = (\star\alpha \wedge \star\star\beta)\omega = (-1)^{k(n-k)}(\star\star\beta \wedge \star\alpha)\omega = (-1)^{k(n-k)} \langle \star\star\beta, \alpha \rangle . \tag{154}$$

Definition 28. *We may define the inner product of* α *and* β *to be*

$$\langle\langle \alpha, \beta \rangle\rangle := \int_M \alpha \wedge \star\beta = \int_M \langle \alpha, \beta \rangle \, \omega. \tag{155}$$

Definition 29. *The codifferential* $\delta : \Omega^k(M) \to \Omega^{k-1}(M)$ *is defined as*

$$\delta = (-1)^{n(k+1)+1} \star d \star . \tag{156}$$

Proposition 18. δ *is the adjoint operator of* d *w.r.t.* $\langle\langle \cdot, \cdot \rangle\rangle$.

Proof. Let $|\alpha| = k-1, |\beta| = k$.

$$\begin{aligned} \int_M d\alpha \wedge \star\beta &= -(-1)^{k-1} \int_M \alpha \wedge d \star \beta = -(-1)^{k-1}(-1)^{(k-1)(n-k+1)} \int_M \alpha \wedge \star\star d \star \beta \\ &= (-1)^{nk+n+1} \int_M \alpha \wedge \star(\star d \star \beta) = \left\langle\left\langle \alpha, (-1)^{n(k+1)+1} \star d \star \beta \right\rangle\right\rangle . \end{aligned} \tag{157}$$

□

Definition 30. *The Laplace-Beltrami operator* $\Delta : \Omega^k(M) \to \Omega^k(M)$ *is defined as*

$$\Delta = d\delta + \delta d. \tag{158}$$

Comment 15. Δ *reduce to* $-\partial_i^2$ *on* $\Omega^0(\mathbb{R}^n)$ *since*

$$\begin{aligned} &(d\delta + \delta d)f = \delta df = (-1)^{n(1+1)+1} \star d \star df \\ &= (-1)^{2n+1} \star d \star \sum_i \partial_i f dx^i = (-1) \star d \sum_i (-1)^{i-1} \partial_i f dx^1 \wedge \cdots \cancel{dx^i} \cdots \wedge dx^n \\ &= (-1)^i \star \sum_j \sum_i \partial_i\partial_j f dx^j \wedge dx^1 \wedge \cdots \cancel{dx^i} \cdots \wedge dx^n = (-1)^{i+i-1} \star \sum_i \partial_i\partial_i f\omega \\ &= -\sum_i \partial_i^2 f. \end{aligned} \tag{159}$$

Proposition 19. *Now we list several properties of Δ:*

$$\begin{aligned}
&1. \quad \Delta\star = \star\Delta. \\
&2. \quad \Delta \text{ is self-adjoint.} \\
&3. \quad \Delta = (d+\delta)^2. \\
&4. \quad \Delta \geq 0. \\
&5. \quad \Delta \text{ commutes with } d \text{ and } \delta.
\end{aligned} \tag{160}$$

Proof.
1.

$$\begin{aligned}
\star d\delta + \star\delta d &= (-1)^{n(k+1)+1} \star d \star d \star + (-1)^{n(k+2)+1} \star \star d \star d \\
&= \delta d \star + (-1)^{n(k+2)+1}(-1)^{k(n-k)} d \star d \\
&= \delta d \star + (-1)^{n(k+2)+1} d(\star d \star)\star \\
&= \delta d \star + (-1)^{n(k+2)+1}(-1)^{n(n-k+1)+1} d\delta\star \\
&= (\delta d + d\delta) \star .
\end{aligned} \tag{161}$$

2.

$$\langle\langle \Delta\alpha, \beta\rangle\rangle = \langle\langle (\delta d + d\delta)\alpha, \beta\rangle\rangle = \langle\langle \alpha, \delta d\beta\rangle\rangle + \langle\langle \alpha, d\delta\beta\rangle\rangle . \tag{162}$$

3.

$$0 = \left\langle\left\langle d^2\alpha, \beta\right\rangle\right\rangle = \left\langle\left\langle \alpha, \delta^2\beta\right\rangle\right\rangle \Rightarrow \delta^2 = 0 \Rightarrow (d+\delta)^2 = d\delta + \delta d. \tag{163}$$

4.

$$\langle\langle \Delta\alpha, \alpha\rangle\rangle = \langle\langle (d+\delta)(d+\delta)\alpha, \alpha\rangle\rangle = \langle\langle (d+\delta)\alpha, (\delta+d)\alpha\rangle\rangle . \tag{164}$$

5. Obvious. □

Definition 31. *α is co-closed if $\delta\alpha = 0$ and co-exact if $\alpha = \delta\beta$. α is harmonic if $\Delta\alpha = 0$.*

Proposition 20. *α is harmonic iff it is closed and co-closed.*

Proof. If α is closed & co-closed, the case is obvious. If α is harmonic,

$$0 = \langle\langle (d+\delta)\alpha, (\delta+d)\alpha\rangle\rangle \Rightarrow (d+\delta)\alpha = 0. \tag{165}$$

So

$$-\langle\langle d\alpha, d\alpha\rangle\rangle = \langle\langle d\alpha, \delta\alpha\rangle\rangle = \left\langle\left\langle d^2\alpha, \alpha\right\rangle\right\rangle = 0. \tag{166}$$

The co-closed case is similar. □

Definition 32. *A weak solution to $\Delta\omega = \alpha$ is a linear functional L on $\Omega^k(M)$ satisfying*

$$L(\Delta\phi) := \langle\langle \alpha, \phi\rangle\rangle . \tag{167}$$

This notion arise from that

$$\langle\langle \omega, \Delta\phi\rangle\rangle = \langle\langle \Delta\omega, \phi\rangle\rangle = \langle\langle \alpha, \phi\rangle\rangle \tag{168}$$

so that $L(\cdot) \sim \langle\langle \omega, \cdot\rangle\rangle$.

Definition 33. *The space of harmonics forms is defined by*

$$\mathcal{H}^k(M) := \{\omega \in \Omega^k(M) : \Delta\omega = 0\}. \tag{169}$$

Theorem 10. *On a closed manifold M,*

$$\Omega^k(M) = \Delta(\Omega^k(M)) \oplus \mathcal{H}^k(M). \tag{170}$$

In other words, any form ω can be uniquely factorized into

$$\omega = d\alpha + \delta\beta + \gamma, \tag{171}$$

where α is co-exact, β is exact and γ is harmonic.

Proof. We want to show that $\Delta(\Omega^k(M)) = \mathcal{H}^k(M)^\perp$. First we show $\Delta(\Omega^k(M)) \subset \mathcal{H}^k(M)^\perp$: Take $\omega = \Delta\eta$ and γ harmonic, then

$$\langle\langle \Delta\eta, \gamma \rangle\rangle = \langle\langle \eta, \Delta\gamma \rangle\rangle = 0. \tag{172}$$

Now we show $\mathcal{H}^k(M)^\perp \subset \Delta(\Omega^k(M))$. This is simple: take $\alpha \in \mathcal{H}^k(M)^\perp$ then we define $L(\Delta\phi) = \langle\langle \alpha, \phi \rangle\rangle$. Extending this to the whole space yields a weak solution. Voilà. To rigorously extend this:

Define l_α on $\Delta(\Omega^k(M)) \subset \Omega^k(M)$ by

$$l_\alpha : l_\alpha(\Delta\phi) = \langle\langle \alpha, \phi \rangle\rangle = \langle\langle \alpha, \phi - P\phi \rangle\rangle \equiv \langle\langle \alpha, \psi \rangle\rangle . \tag{173}$$

There is nothing deep but to let ϕ be the in the orthogonal complement by projection.

Lemma 9.

$$\exists c > 0 : \forall \alpha \in \mathcal{H}^k(M)^\perp, ||\alpha|| \leq c||\Delta\alpha||. \tag{174}$$

Proof. Assume the contrary. Then $\forall c_i > 0, \exists \alpha_i \in \mathcal{H}^k(M)^\perp : ||\alpha_i|| > c_i||\Delta\alpha_i||$. WLOG take $||\alpha_i|| = 1$, since the norm does not affect the inequality. Take $c_i = i$ then $\{\Delta\alpha_i\}$ is a sequence whose norm goes to 0. Since $||\Delta\alpha_i|| \leq 1$, we have this sequence being Cauchy. So we are able to define the following functional L for this certain sequence:

$$L(\psi) := \lim_{i\to\infty} \langle\langle \alpha_i, \psi \rangle\rangle . \tag{175}$$

Since L is bounded, we have

$$L(\Delta\phi) = \lim_{i\to\infty} \langle\langle \alpha_i, \Delta\phi \rangle\rangle = \lim_{i\to\infty} \langle\langle \Delta\alpha_i, \phi \rangle\rangle = \left\langle\left\langle \lim_{i\to\infty} \Delta\alpha_i, \phi \right\rangle\right\rangle = 0. \tag{176}$$

Hence α, the limit of α_i is a solution to $\Delta\alpha = 0$. So $\alpha = 0$ and it does not have norm 1. Hence the contradiction. □

So

$$||l_\alpha(\Delta\phi)|| = || \langle\langle \alpha, \psi \rangle\rangle || \leq c||\alpha|| \cdot ||\Delta\psi||. \tag{177}$$

Since $\Delta P\phi = 0$,

$$||\Delta\psi|| = ||\Delta\phi|| \Rightarrow ||l_\alpha(\Delta\phi)|| \leq c||\alpha|| \cdot ||\Delta\phi||. \tag{178}$$

Lemma 10 (Hahn–Banach theorem)**.** *If on a subspace $U \subset V$ we have a bounded linear functional l, then it can be extended on V.*

Since l_α is bounded, it can be extended on $\Omega^k(M)$ to L_α. Hence L_α is a weak solution and there exists ω such that $\Delta\omega = \alpha$. □

Comment 16. *This resembles the fundamental theorem of linear algebra, for the first space is the image, and the second one the kernel.*

Comment 17. *Consider the Laplace equation $\Delta\omega = \alpha$. It is solvable iff $\alpha \in \Delta(\Omega^k(M))$. In other words, $\alpha \in \mathcal{H}^k(M)^\perp$. In fact, this is already presented in the main proof above.*

Definition 34. *Let the Green's operator G be defined as*

$$G : \Omega^k \to (\mathcal{H}^k)^\perp \tag{179}$$

to be the unique solution to

$$\Delta(G\alpha) = \alpha - P\alpha. \tag{180}$$

The solvability is guaranteed by resrtricing $\alpha - P\alpha$ to be in $(\mathcal{H}^k)^\perp = \Delta(\Omega^k)$. The uniqueness is guaranteed by restricting G to $(\mathcal{H}^k)^\perp$.

Proposition 21. *G commutes with d, δ and hence Δ.*

Proof.

$$\Delta(dG\alpha - Gd\alpha) = d\Delta G\alpha - (d\alpha - Pd\alpha) = Pd\alpha - dP\alpha. \tag{181}$$

Since $\alpha = d\mu + \delta\nu + \gamma$, $dP\alpha = d\gamma = 0$ and $Pd\alpha = Pd\delta\nu = 0$ since $\langle\langle d\delta\nu, \omega \rangle\rangle = 0$ for any harmonic ω. So the difference is 0 and the case for δ is similar. □

Theorem 11. *Any cohomology class $[\omega] \in H^k(M)$ can be represented by a unique harmonic form.*

Proof. Let $\omega \in \Omega^k(M)$,

$$\begin{aligned} \omega &= d\delta G\omega + \delta dG\omega + P\omega \\ &= d\delta G\omega + \delta Gd\omega + P\omega. \end{aligned} \tag{182}$$

If $\omega \in Z^k(M)$, $d\omega = 0$ so the representation of $[\omega]$ is in the form of

$$d\delta G\omega + P\omega + d\eta. \tag{183}$$

So $P\omega$ does represent $[\omega]$ and is harmonic. Now if α_1, α_2 are harmonic,

$$\langle\langle \alpha_1 - \alpha_2, \alpha_1 - \alpha_2 \rangle\rangle = \langle\langle du, \alpha_1 - \alpha_2 \rangle\rangle = \langle\langle u, \delta\alpha_1 - \delta\alpha_2 \rangle\rangle = \langle\langle u, 0 \rangle\rangle = 0 \tag{184}$$

since harmonic forms are closed and co-closed. □

Corollary 8. $H^k(M) \simeq \mathcal{H}^k(M)$.

Proposition 22. *The unique harmonic representation is of minimal norm among all representations.*

Proof.

$$\langle\langle \omega + d\eta, \omega + d\eta \rangle\rangle = ||\omega||^2 + ||d\eta||^2 + 2\cancel{\langle\langle \delta\omega, \eta \rangle\rangle} = ||\omega||^2 + ||d\eta||^2. \tag{185}$$

□